TECHNICAL REPORT

CASCADE: predicting shipment ETA and explaining where delays come from

Aiqon AI Internship take-home · model v1.2.0 · 2026-10-13

Summary. CASCADE predicts when a container shipment will actually be delivered, how likely it is to be more than 24 hours late, and which stage of the journey is causing the delay. On a test season it never saw in training, its delivery-time error is **24.5 h**, which is **37% lower** than assuming today's delay stays the same. Because our simulated data records the true cause of every hour of delay, we could check the explanations too: the real cause is in CASCADE's top two stages **68%** of the time.

Table 1 · Key results on the held-out test season

What we measured	Why it matters	Result	Average over 5 runs
Error reduction vs carry-forward	Gain over "today's delay stays the same"	**37%**	28% ± 6%
Error reduction vs propagation prior	Gain over a lane-level delay propagation rule	**30%**	22% ± 7%
P10–P90 range coverage	How often the real delay falls in the predicted range (aim 80%)	**81%**	81% ± 2%
Chance-late accuracy (PR-AUC)	How well P(late > 24 h) separates late from on-time shipments	**0.85**	0.87 ± 0.02
Right cause, first choice	Explanation checked against the true cause	**42%**	47% ± 4%
Response time (p95)	Time to predict and explain one shipment	**1.8 ms**	

1 · How ETA and delay are defined

Every shipment passes through the same stages: gate-in, origin departure, hub arrival and departure (only on hub lanes), destination arrival, discharge, customs release, gate-out and delivery. Each stage has a planned time and an actual time. All values are in hours, and a positive number means late.

`Stage delay`	actual time − planned time at that stage
`Stage increment`	how much delay a stage adds (or recovers) compared with the stage before it
`Final delay`	delay at delivery, which equals the sum of all stage increments
`ETA`	planned delivery + delay so far + predicted remaining delay, given as P10, P50 and P90
`Late`	final delay greater than 24 hours

At each milestone the model only sees what was known at that moment. Congestion and weather for future stages are given as forecasts, not as the values that actually happened.

2 · Data

We generated 300 shipments with a simulator built on real logistics rules, on lanes from Asian ports through hubs such as Singapore and Colombo to Europe and the US. Each rule in Table 2 writes the hours of delay it causes into a **cause ledger**, which gives us the true reason for every delay. Some milestone reports are deliberately missing or arrive late, as they do with real carrier data. The last voyages in time, which fall in the busy Q4 peak, are held out as the test set.

Table 2 · Real-world behaviour built into the simulator

Stage	What happens
Origin	Carrier reliability, port congestion and storms delay departure.
Ocean	Bad weather slows the vessel; ships speed up to recover part of a delay.
Transshipment	If the container misses its connection, it waits for the next sailing. This is the biggest source of cascading delay.
Destination port	A late vessel loses its berth slot and queues longer when the port is busy.
Customs	Inspections depend on cargo risk and paperwork; release only happens in office hours.
Inland	Trains leave at fixed times, trucks need appointments and receivers have delivery windows.

3 · Method

Table 3 · How CASCADE builds a prediction

Step	What it does
Snapshots	One row per shipment per milestone, using only information available at that time.
Operator rulebook	Projects the delay each remaining stage will add, using cut-offs, sailing schedules, queues and calendars. Its settings are deliberately different from the simulator's.
Stage features	Link stages together: hub buffer vs connection cut-off, congestion × current lateness, time to the next customs slot.
Machine learning	LightGBM quantile models give the P10–P90 range, a calibrated classifier gives P(late), and conformal calibration keeps the range reliable when the season changes.
Explanation	The prediction is split into delay already realised plus each remaining stage's share, and written as a short sentence for the customer or the planner.

On validation data, the learned correction to the median did not beat the rulebook, so it was switched off (weight 0.0). The median ETA therefore comes from the rulebook, and machine learning adds the range, the chance of being late and the stage forecasts.

4 · Results

Table 4 · Delivery-time error (725 test predictions)

Method	MAE	RMSE
Assume the original plan holds	56.3 h	88.6 h
Assume today's delay stays the same	38.6 h	64.5 h
Lane-level propagation prior	35.0 h	59.1 h
CASCADE	**24.5 h**	**50.8 h**

Table 5 · Is the stated range reliable?

Shipment position	Coverage	Range width
Pre-departure	82%	6.8 days
At sea	74%	3.9 days
At port	84%	2.3 days
Post-customs	91%	11 h
Overall	**81%**	**3.5 days**

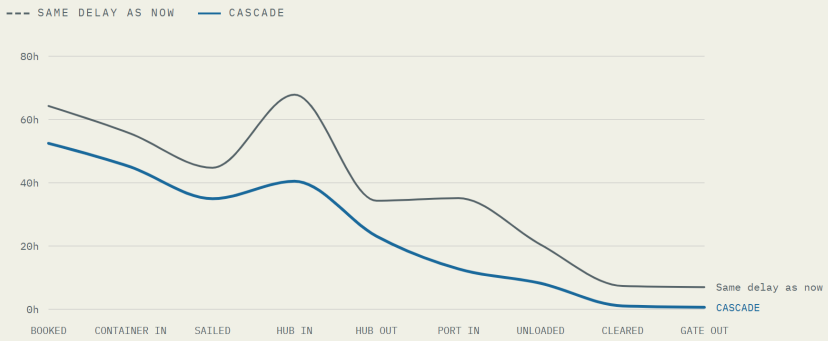


Figure 1 · Prediction error at each milestone, from the dashboard. Error falls from 52.5 h at booking to 0.6 h at gate-out. The jump at the hub is where connections are made or missed.

Table 6 · Does it blame the right stage?

Method	Correct
CASCADE, top two stages	**68%**
CASCADE, first choice	42%
Lane-history guess	26%
Random guess	18%

Table 7 · How much one hour of delay grows by delivery

Delay added at	CASCADE	True
Destination arrival	1.23	1.58
Hub arrival	1.79	2.39
Origin departure	1.67	1.90
Discharge	0.82	0.76

Table 7: the true values come from re-running the simulator with 12 extra hours injected at one stage. A value above 1 means the delay grows on the way to the customer.

Worked example: SHP-000174, Shanghai → Tanjung Pelepas → Rotterdam

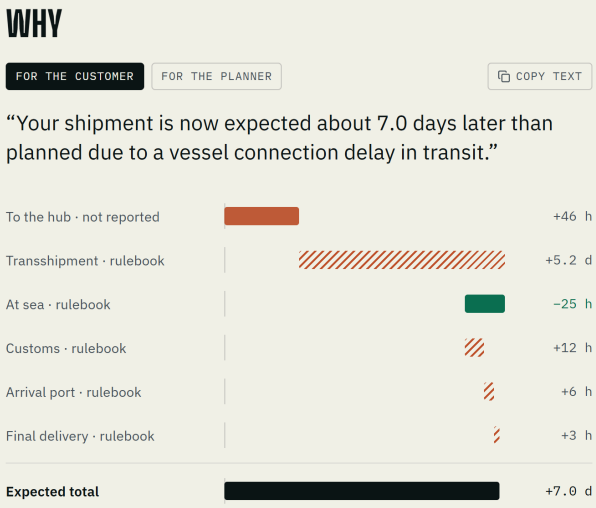


Figure 2 · The dashboard's explanation for this shipment at hub arrival. Delay so far (solid) + projected stage delay (hatched) = P50 (168 h).

> SHP-000174 is expected 7.0 days late (likely 4.3 days–10.4 days late); 83% chance of more than 24 h late. Already 46 h behind plan. Main driver: Transshipment — projected to reach Tanjung Pelepas after the connection cut-off (buffer short by 36 h) (+126 h). Also contributing: Customs — customs release projected to fall outside office hours / on a non-working day at Rotterdam (+12 h). Suggested action: Check the next sailing from the hub and consider re-booking or priority loading. Confidence is reduced: 2 milestone(s) missing from the feed.

The biggest driver is a projected missed connection at the hub (126 h). The simulator's cause ledger confirms the true cause: **Transshipment**. Expected cost of the delay: $778.